

Sibling Spillovers by Education: Evidence from U.S. Compulsory Schooling and Educational Expansion Reforms

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Abstract

Education plays an important role in success in the labor market and beyond. This study investigates whether siblings' education affects each other's labor market outcomes, or, in other words, sibling spillovers of education. Isolating peer effects is a challenge, as siblings' socioeconomic outcomes are jointly determined by their shared family and overall environment. To tackle this question, I use exogenous variation induced by historical compulsory schooling and educational expansion reforms. I use linked, full-population U.S. census data (1850-1940) and data on the timing of reforms in U.S. states in a staggered Difference-in-Differences design. In a first stage, I show that the reforms had positive effects on human capital, measured as literacy and educational attainment. Subsequently, I show that these increases in human capital had positive spillover effects on occupational earnings from older to younger siblings. There are no spillover effects from younger to older siblings. In other words, individuals choose "better" occupations, meaning occupations with higher prestige and pay, when their next older sibling gains more education, but not when their next younger sibling gains more education.

Introduction

Families are important actors in the production of inequality (McLanahan & Percheski, 2008; Hansen & Toft, 2021; Maralani & Portier, 2021; Silverman et al., 2024). While much of the social stratification literature is concerned with the intergenerational transmission of (dis)advantage, from parents to children, a more recent literature also investigates intragenerational transmission, that is, from sibling to sibling (e.g., Zang et al., 2023). Siblings can be each other's most important peers as they spend extensive amounts of time with each other in childhood, more than they do with parents (Dunifon et al., 2017), and will be each other's longest-living kin (White, 2001; Connidis & Barnett, 2018). They can thus be important both early on in life, before individuals find a partner or have children of their own, and later on in life once parents and other kin have passed away.

That siblings therefore affect each other’s life outcomes in a meaningful way may be easy to argue, given their assumed presence in each other’s lives; however, identifying effects of one sibling on another remains difficult. Siblings, like all peers, are embedded in the same social contexts, sharing their parents and the neighborhoods they grow up in. Isolating spillover effects - effects of one sibling on another beyond other shared factors - can be important to understand mechanisms of status transmission and persistence, as well as the role families play in this process. I use a historical case, in which compulsory schooling reforms exogenously changed educational attainment for individuals, to investigate whether this increase in human capital had a positive effect on the occupational status of their siblings. The question of whether not only one’s own, but also one’s sibling’s, human capital matters for socioeconomic outcomes has important implications for inequality dynamics and policy implementation.

Data & Method

I use linked U.S. decennial full-population census data from 1850 to 1940 (Buckles et al., 2023). For identification, I link these data with codings of the full universe of compulsory schooling and educational expansion legislation for each federal state (which I have harmonized and extended based on Goldin and Katz (2008a), Clay et al. (2021), and Knapp et al. (2025)). My analytical sample includes ~287 million observations from ~90 million individuals, who have ~72 million older siblings and ~73 million younger siblings.

I use a staggered Difference-in-Differences design, as the federal states enacted the same compulsory schooling and educational expansion reforms at different times. I use the estimator by Callaway and Sant’Anna (2021) to estimate event studies for repeated cross-sectional data to account for treatment effect heterogeneity (cf. Rüttenauer & Aksoy, 2024; Baker et al., 2025).

I proceed in two steps. To first show that the reforms were effective in raising human capital and education, I estimate a first stage on an individual’s literacy (being able to read and write) and educational attainment (years of schooling). In a second step, I estimate sibling spillover effects, i.e., how this increase in human capital spills over from the next older or next younger sibling. I estimate spillovers on occupational earnings, an index that ranges from 0-100 and assigns each person the median earnings of their occupation in comparison to all other workers. For example, a nurse has an occupational earnings score of 40, meaning that a nurse earns more than 40% of all other workers. Practically, in the first stage, I use an individual’s birth cohort to define their treatment status. In the second step, I use their next older or next younger sibling’s birth cohort to define treatment status.

Preliminary Results

For all results, I show 10 pre-treatment birth cohorts to provide evidence for parallel trends and 11 post-treatment cohorts to show whether effects phase in and wear off. Birth cohort 0 on the x-axis is always the first treated cohort. I begin by showing first-stage results. Results for literacy are shown in Figure 1a and for educational attainment in Figure 1b.

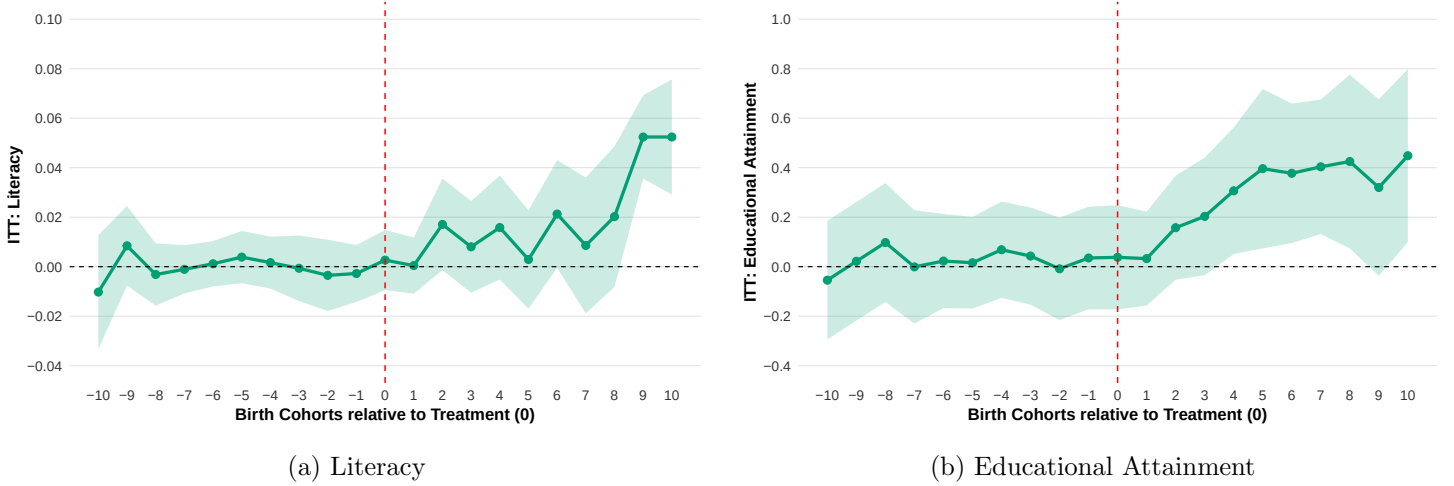
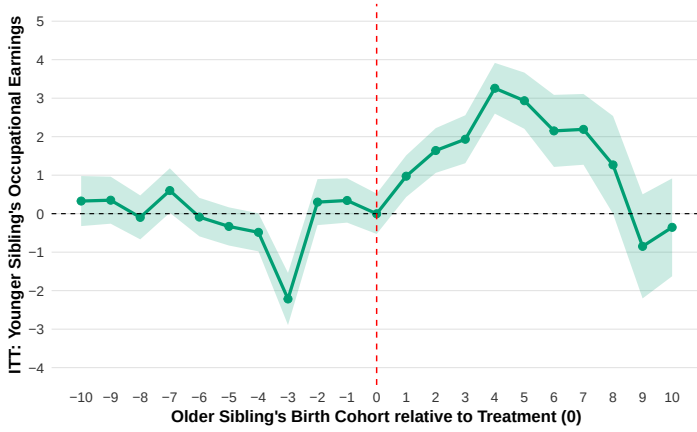


Figure 1: First Stage Effects of Schooling Reform, Event Study Estimates

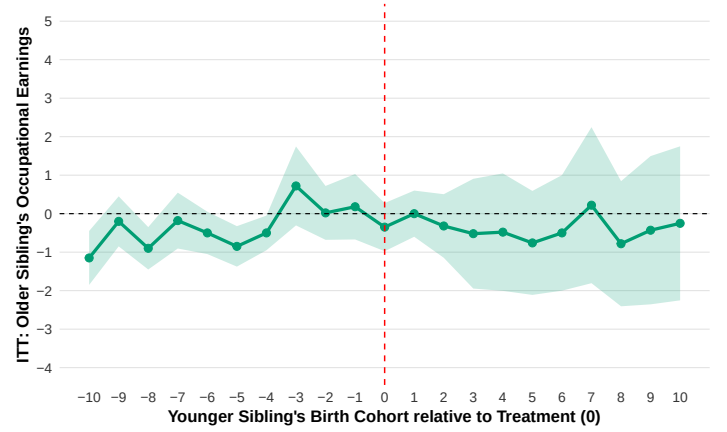
Results for literacy suggest that the initial introduction of compulsory schooling had small positive effects on literacy, but the effects for the initial treated cohorts are extremely small, while only in later cohorts do the effects seem to “take off.” Results for educational attainment look somewhat similar. Overall, treatment effects remain initially small and only take off after a few treated cohorts. This provides evidence that the reforms were effective in raising the human capital of affected individuals. Now I turn to the question of whether this increase in human capital also spills over to the siblings of the affected individuals.

In Figure 2a, I turn to event study estimates of the closest older sibling receiving treatment on an individual’s occupational earnings. In this specification I crucially control for the individual’s birth cohort (i.e., the individual who receives the spillover) to control for the direct effect of being compelled to go to school. Results for the older to younger spillover seem to suggest positive effects that “take off” for the first few cohorts and quickly phase out thereafter. In Figure 2b, I show event study estimates of the closest younger sibling receiving treatment. Results for these spillovers remain effectively null, with some of the later cohorts having negative point estimates, although individual cohort estimates can remain somewhat volatile and are not statistically significant.

In total, these results suggest that the reforms did boost human capital on the population level. Positive sibling spillover effects can be seen for spillovers from older to younger siblings, but not from younger to older siblings.



(a) Older to Younger



(b) Younger to Older

Figure 2: Sibling Spillover Effects on Occupational Earnings, Event Study Estimates

Discussion

In the full paper, I discuss implications for theories of human capital accumulation, education as positional good, and the role of families for inequality. For example, if sibling spillovers are more important for worse-off families, i.e., as in Zang et al. (2023), overall inequality may decrease as a function of sibling spillovers.

Ultimately, the insights from this project also speak to modern policy debates, policy implementation (Figlio et al., 2023), and renewed calls for educational expansion (cf. Goldin & Katz, 2008b; van de Werfhorst, 2024). In lower- and middle-income countries, basic schooling is still not universal. In wealthier contexts, the challenge has shifted to higher education, raising enrollment and completion rates at the tertiary level. Understanding how past education reforms created ripple effects within families can inform more effective education policies today, both in expanding opportunity and in accounting for the broader social impact of schooling in varying contexts across the globe. Results from this study suggest that expanding education benefits not only those directly affected by it but also benefits their close peers in the family.

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